Lab 1.1

Ringing in Cables and Scope Terminations

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Setup

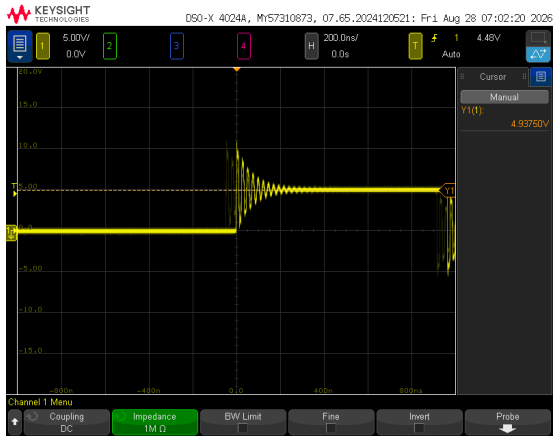
- Scope
  - Keysight DSOX4024A, 200 MHz, 8-bit vertical resolution
  - Input impedance switched between 1 MΩ and 50 Ω
  - Averaging off for set up, then on to remove digitizing noise
- Cable and adapters
  - 3 ft RG58 coax (Z_0 = 50 Ω), SMA-to-BNC adapter
  - RG58 specs: velocity factor of 0.66 → 1.54 ns/ft → TD ≈ 4.6 ns for 3 ft of cable alone
- Source:
  - 555 timer driving a hex inverter at 5 V
  - Series output resistor: 3 Ω, 50 Ω, 950 Ω
- Six configurations: 3 sources and 2 scope terminations

Thevenin Source Model

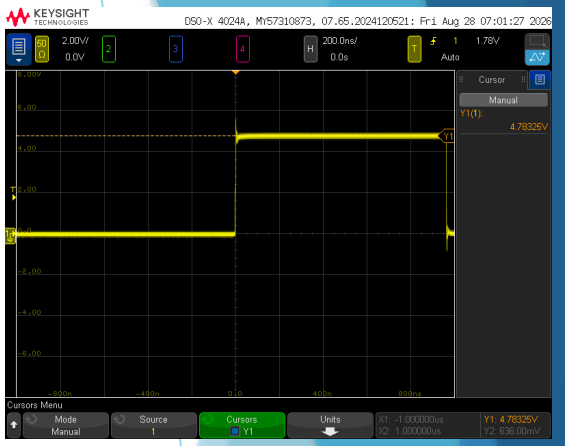
- Open circuit, 1 MΩ, steady state: $V_L = V_{th}$
- Loaded, 50 Ω, steady state: $V_L = 50 \times V_{th} / (R_{th} + 50)$
- Solve for the source resistance: $R_{th} = 50 \times (V_{th} / V_L - 1)$
- 3 Ω source: $R_{th} = 50 \times (4.94 / 4.78 - 1) = 1.7\ \Omega$

Source	V_{th} (1 MΩ)	V_L (50 Ω)	R_{th} measured	$V_{launched}$
3 Ω	4.94 V	4.78 V	1.7 Ω	4.78 V
50 Ω	5.00 V	2.53 V	49 Ω	2.53 V
950 Ω	5.03 V	0.50 V	455 Ω	0.50 V

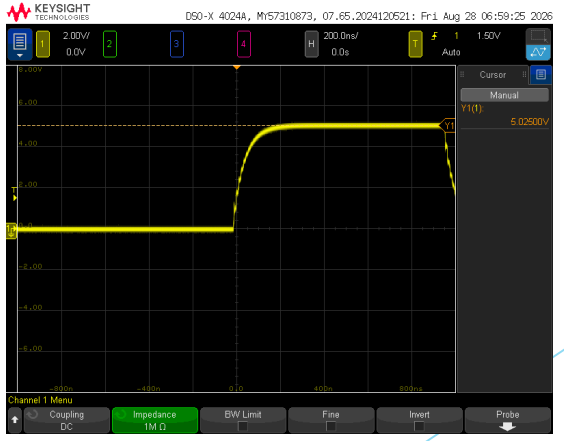
- V_{th} ≈ 5.0 V for all three: same inverter output with only the series resistor changing
- 3 Ω value is scope-limited: at 5 V/div one LSB is 0.16 V, so $R_{th} = 1.7 \pm 1.7\ \Omega$
- The “950 Ω” position measures 455 Ω. The step ratio on slide 6 gives ≈ 470 Ω independently, so it is the resistor, not the measurement method.
- The 10.9 V first peak on the 3 Ω / 1 MΩ shot is NOT V_{th}.



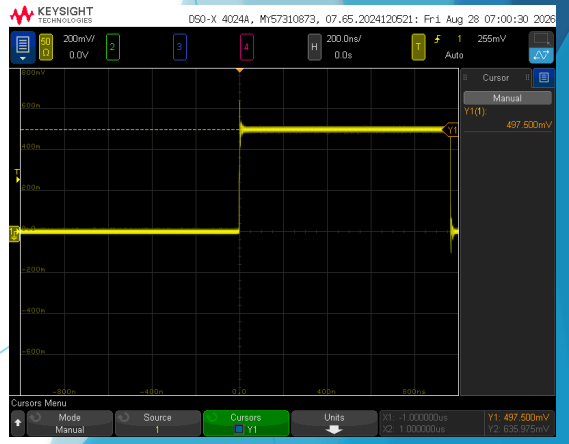
3 Ω → 1 MΩ: flat top = V_{th} = 4.94 V



3 Ω → 50 Ω: V_L = 4.78 V



950 Ω → 1 MΩ: V_{th} = 5.03 V

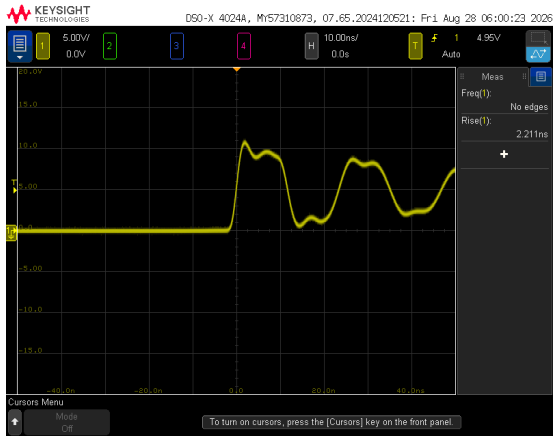


950 Ω → 50 Ω: V_L = 0.50 V

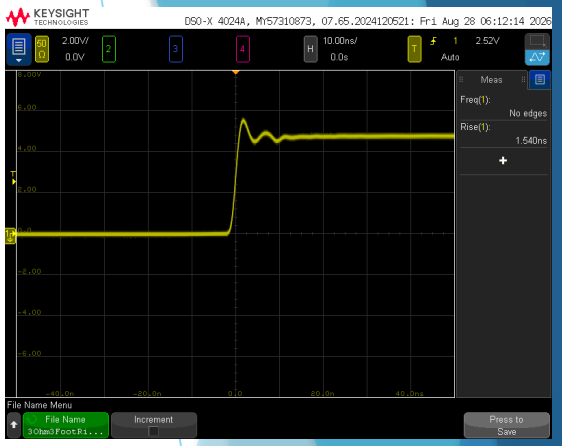
3 Ω Source: Reflections Dominate

- **Expected:** ρ_{src} = (1.7 − 50)/(1.7 + 50) = -0.93 and ρ_{scope} = +1 at 1 MΩ → large ringing with period 4 x TD, decaying by ρ_{src}^2 ≈ 0.87 each period
- $V_{launched}$ = 4.78 V doubles at the open scope → 9.6 V. The source itself overshoots to 5.56 V (seen on 50 Ω), and 2 × 5.56 = 11.1 V vs 10.9 V measured
- Each round trip adds 2 x $V_{launched\ x}\ \rho_{src}^n$: **9.6 → 0.6 → 9.0 → 1.2 V**
- **Period:** cable alone predicts 4 × 4.6 = 18.5 ns; measured 24.3 ns → effective TD ≈ 6.1 ns. The extra ≈ 1.5 ns is adapters, board trace and the scope's ~15 pF input
- **Into 50 Ω:** ρ_{scope} = 0, nothing reflects → flat 4.78 V

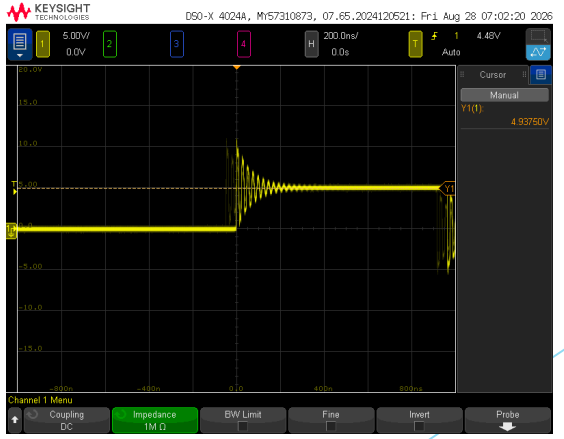
3 Ω / 1 MΩ feature	Predicted	Measured
1st peak	9.6 V (11.1 V with overshoot)	10.9 V
1st trough	0.6 V	0.5 V
2nd peak	9.0 V	8.9 V
2nd trough	1.2 V	1.8 V
Ringing period (4·TD)	18.5 ns (cable only)	24.3 ns → TD ≈ 6.1 ns
Steady state	4.94 V	4.94 V



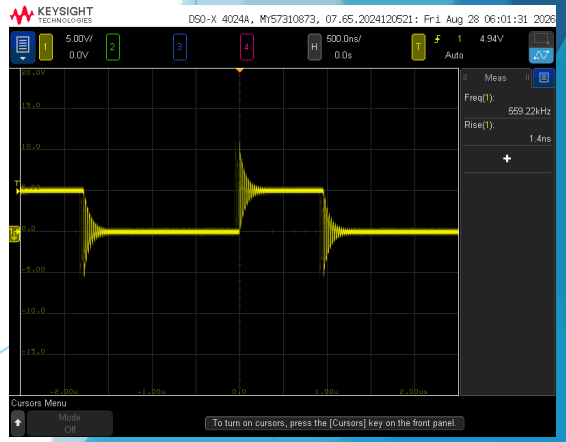
3 Ω → 1 MΩ, 10 ns/div: period 24 ns



3 Ω → 50 Ω, 10 ns/div: flat 4.78 V



3 Ω → 1 MΩ, 200 ns/div: slow decay

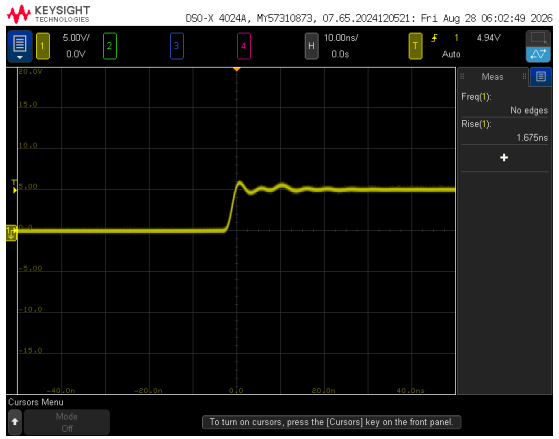


3 Ω → 1 MΩ, 500 ns/div: both edges ring

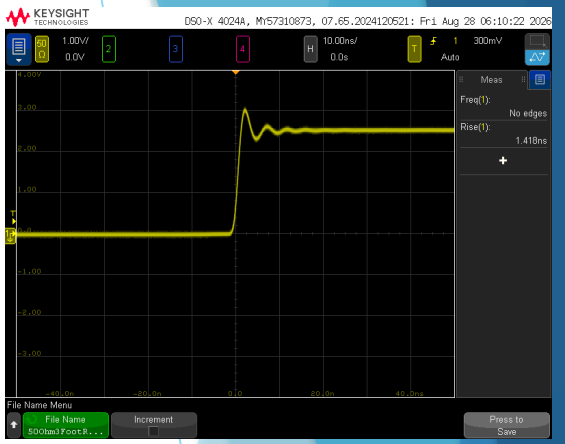
50 Ω Source: Matched, No Ringing

- **Expected:** R_{th} = 49 Ω → ρ_{src} = -0.01. Anything the scope reflects is absorbed when it gets back to the source
- **Into 1 MΩ:** $V_{launched}$ = 2.5 V doubles at the open end to 5.0 V after one TD, and it is done: a single step, no ringing
- **Into 50 Ω:** 2.5 V is absorbed at the scope → flat 2.5 V, half of V_{th}, exactly the voltage-divider result
- The same ~5 ns wiggle appears on both terminations, **so it cannot be a cable reflection (slide 7)**

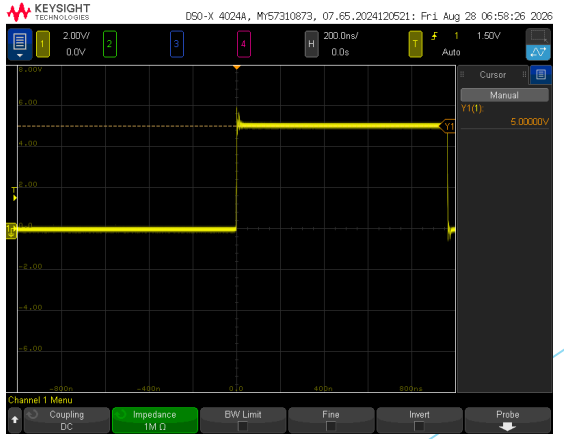
50 Ω source	Predicted	Measured
1 MΩ steady state	5.00 V	5.00 V
50 Ω steady state	2.50 V	2.53 V (+1%)
Reflection ringing at 4·TD	none (ρ_{src} ≈ 0)	none
10–90 rise into 50 Ω	scope-limited	1.42 ns



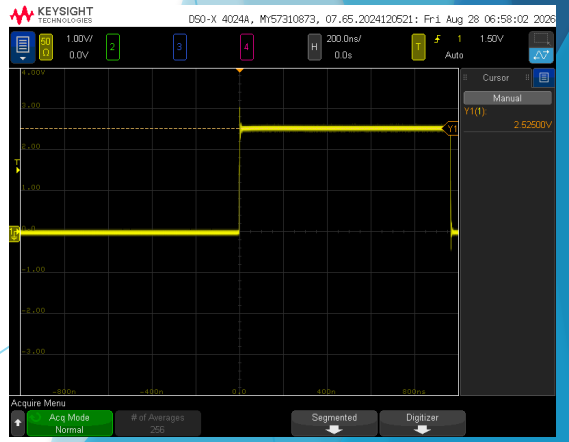
50 Ω → 1 MΩ, 10 ns/div: one 5 V step



50 Ω → 50 Ω, 10 ns/div: flat 2.5 V



Steady state, 200 ns/div: 5.00 V

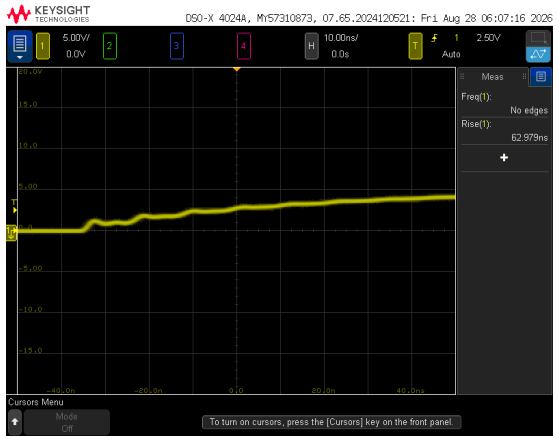


Steady state, 200 ns/div: 2.525 V

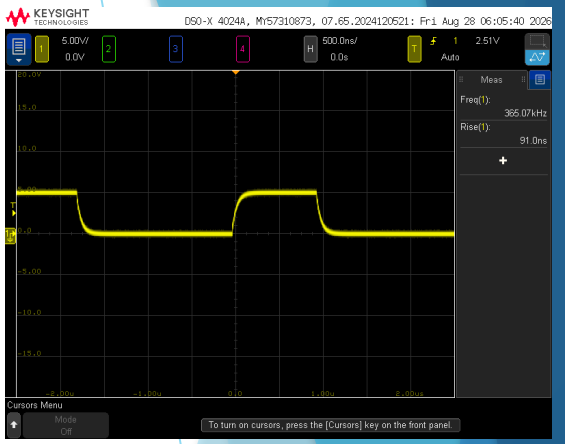
950 Ω Source: Slow Stepped Rise

- **Expected:** ρ_{src} = (455 − 50)/(455 + 50) = +0.80 and ρ_{scope} = +1 → each round trip adds a step 0.80x the last one
- First step = 2 x $V_{launched}$ = 2 × 0.50 = 1.0 V, then a new step every 2xTD ≈ 12 ns (using TD from the 3 Ω ringing)
- Overall rise ≈ R x C with C ≈ 91 pF of cable + 15 pF scope: τ ≈ 48 ns → 10–90 ≈ 105 ns
- Step ratio read from the CSV is 0.80 to 0.81 → R_{th} ≈ 470 Ω (agreeing with 455 Ω from the 50 Ω load). **The "950 Ω" setting is really ≈ 460 Ω**
- **Into 50 Ω:** flat 0.50 V (a true 950 Ω would give 0.25 V)

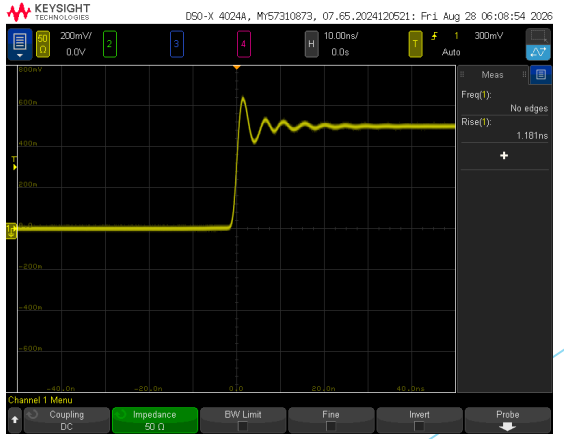
950 Ω source	Predicted (R_{th} = 455 Ω)	Measured
1 MΩ first step	1.0 V	0.99 V
1 MΩ step spacing (2·TD)	9.2 ns (cable only) – 12 ns	10–12 ns
1 MΩ 10–90 rise (2.2·τ)	≈ 105 ns	91 ns
50 Ω steady state	0.50 V	0.50 V



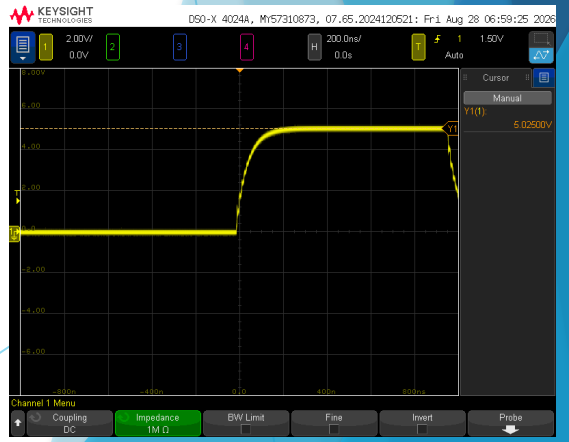
950 Ω → 1 MΩ, 10 ns/div: steps 2·TD apart



950 Ω → 1 MΩ, 500 ns/div: 91 ns rise



950 Ω → 50 Ω, 10 ns/div: 0.50 V flat



950 Ω → 1 MΩ, 200 ns/div: 5.03 V

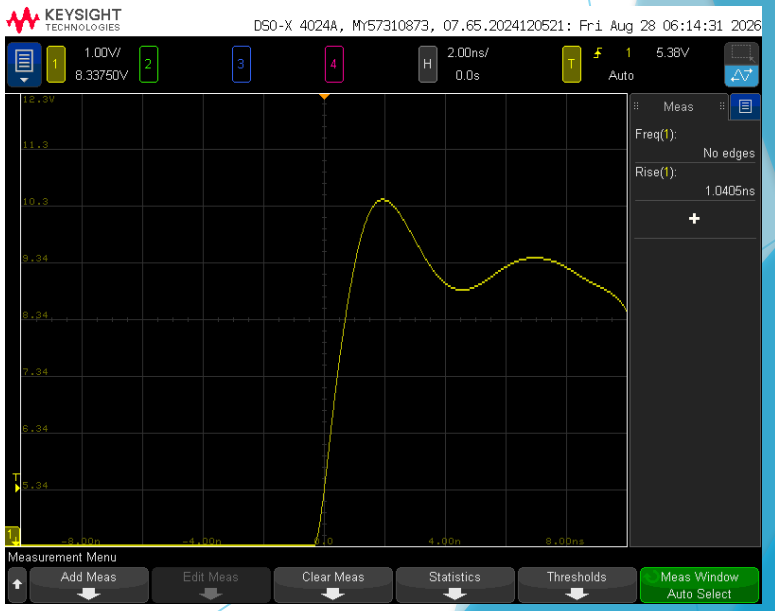
Why Ringing Remains on 50 Ω

- **Observed:** a ~5 ns (~200 MHz) ring, about 3 cycles, on every 50 Ω capture, independent of source resistance
- **If it were cable reflections** the period would be 4 x TD ≈ 24 ns (or 2 x TD ≈ 12 ns). 5 ns does not fit
- A 50 ± 3 Ω cable/scope mismatch gives ρ ≈ 0.03 → ≤ 3% at the wrong period. We see 15–25% overshoot
  - ρ = (R_scope − Z0) / (R_scope + Z0) = (50 - 53) / (50 + 53) ≈ -0.03
- It also appears with the 50 Ω source into 1 MΩ, where ρ_{src} ≈ 0 kills reflections after one round trip
- **Conclusion:** We are watching rail ringing
- **Next check:** probe the quiet-HIGH inverter output with the 10x probe. If the same ~5 ns ring appears there, time-aligned with the switching edge, it is the rail

Rise Time and So What?

- **Three-parameter source model**
  - V_{th} ≈ 5.0 V (all three settings)
  - R_{th} = 1.7 Ω / 49 Ω / 455 Ω (nominal 3 / 50 / 950 Ω)
  - Rise time ≤ 1.2 ns
- **Intrinsic or not?** V_{th} and R_{th} are intrinsic. The rise time and shape we record are not since they depend on termination, TD, and scope bandwidth.
- **Shortest rise time we can measure:** the 200 MHz scope's own ≈ 1.75 ns (0.35 / BW). Our 1.2 to 1.5 ns readings sit near that level so we know that the source is at least that fast.
- **So what?**
  - Low-impedance source + 1 MΩ input = reflection ringing. Use 50 Ω to remove it
  - You have to be careful with 50 Ω since the scope will see 0.5 V instead of 5V (with the 950 Ω source) and is limited to 5 Vrms.
    - (Measuring something like a 12V rail can ruin the scope's resistor)
    - 50 Ω is good for reducing reflections but can change how the circuit behaves

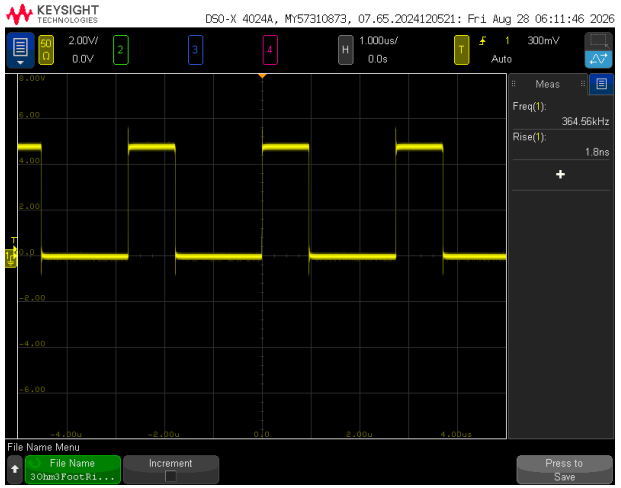
10–90 rise into 50 Ω	Measured
3 Ω source	1.54 ns
50 Ω source	1.42 ns
950 Ω source	1.18 ns
Scope spec, 200 MHz	≈ 1.75 ns



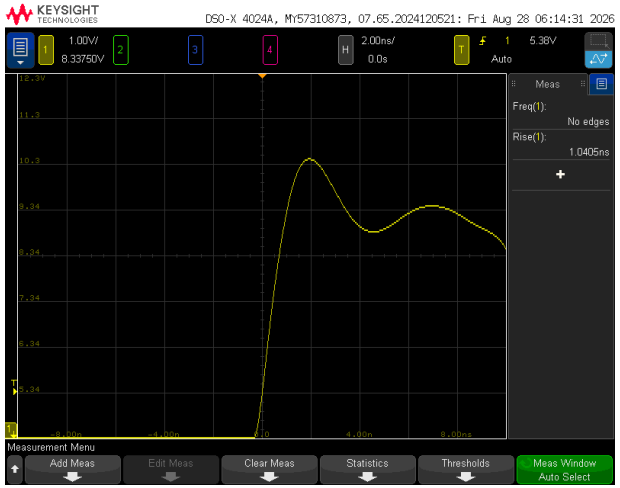
3 Ω → 1 MΩ, averaged, 2 ns/div: edge 1.04 ns, digitizing noise gone

Questions?

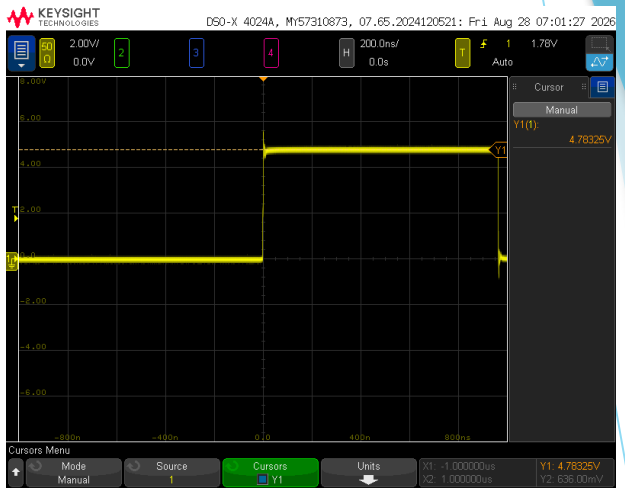
Additional scope captures



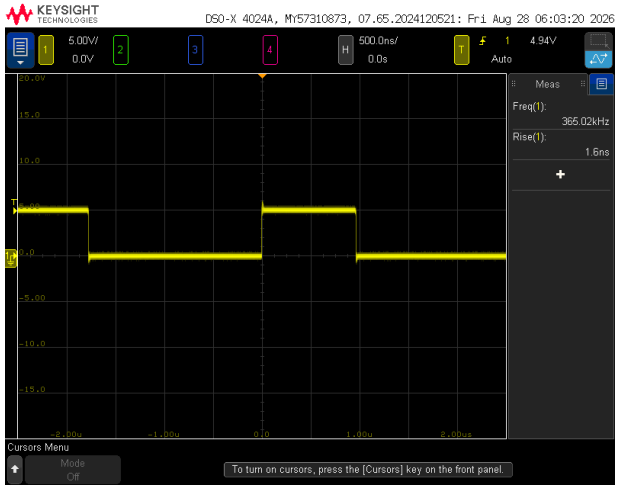
3 Ω → 50 Ω, 1 µs/div



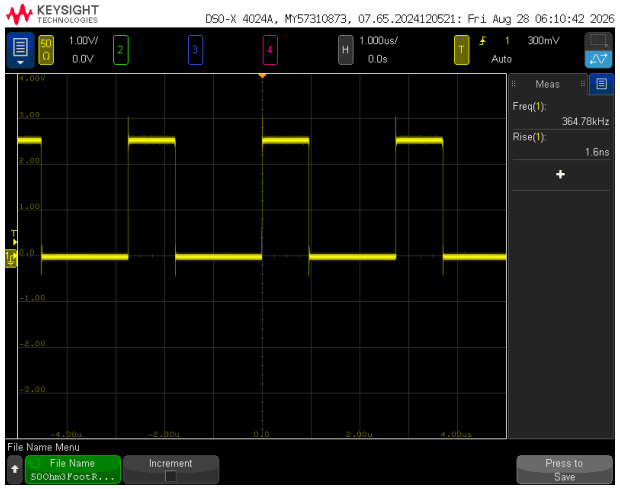
3 Ω → 1 MΩ first peak, averaged



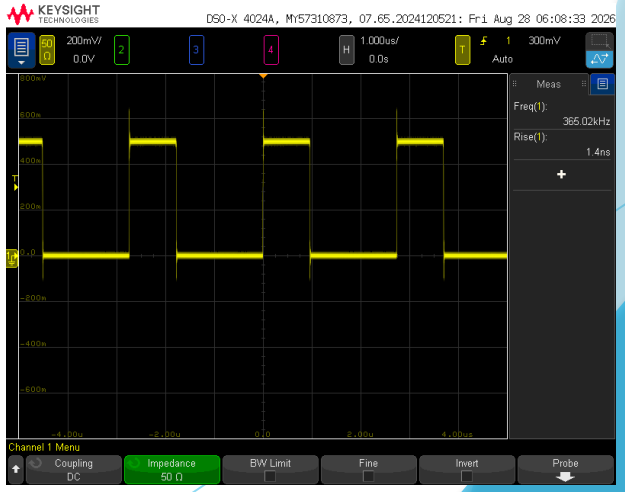
3 Ω → 50 Ω, 200 ns/div



50 Ω → 1 MΩ, 500 ns/div



50 Ω → 50 Ω, 1 µs/div



950 Ω → 50 Ω, 1 µs/div